[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

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Class 3 Maths — CA-1 (Hard) — Answer Key

Class : 3	Max Marks : 70
Subject : Maths (Math-Mela)	Duration : 2 Hr 30 Min

Syllabus: Chapters 1–3 — What's in a Name?, Toy Joy, Double Century

General Instructions: All questions are compulsory. For Section A, write the correct option letter.

ANSWER KEY

SECTION A — Multiple Choice Questions (1 Mark each)

1. A basket has 187 mangoes. If they are packed into boxes of 10, how many full boxes can be made and how many mangoes are left over? *(1 Mark)*

A. 17 boxes, 7 left over
B. 18 boxes, 7 left over ✓
C. 18 boxes, 0 left over
D. 19 boxes, 7 left over

2. Which shape, when cut straight through the middle parallel to its base, always shows a circle as the cut surface? *(1 Mark)*

A. Cube
B. Cuboid
C. Cylinder ✓
D. None of these

3. A cuboid has length, breadth, and height all different from each other. How many pairs of identical (equal) faces does it have? *(1 Mark)*

A. 1
B. 2
C. 3 ✓
D. 6

4. If the number 1_5 (tens digit missing) is greater than 143 but less than 168, which digit could go in the blank? *(1 Mark)*

A. 3
B. 5 ✓
C. 2
D. 9

5. What is the greatest 3-digit number you can make that is less than 200 using the digits 1, 9, and 8 (using each digit exactly once)? *(1 Mark)*

A. 198 ✓
B. 189
C. 918
D. 981

6. A toy is made by joining a cone on top of a cylinder. How many total flat faces does the combined toy have (counting only the outer visible flat faces)? *(1 Mark)*

A. 1
B. 2 ✓
C. 3
D. 4

7. Which of these statements is TRUE about all cubes? *(1 Mark)*

A. A cube is never also a cuboid
B. Every cube is a special type of cuboid ✓
C. A cube has curved surfaces
D. A cube has only 4 faces

8. The sum of the hundreds digit and tens digit of a number is 3, and the number is between 100 and 200. Which of these could the number be? *(1 Mark)*

A. 121 ✓
B. 300
C. 213
D. 450

9. If you remove one flat face from a cube by cutting it open flat (unfolding), how many flat pieces (faces) would you see in total in the unfolded net? *(1 Mark)*

A. 4
B. 5
C. 6 ✓
D. 8

SECTION B — Short Answer Questions (2 Marks each)

1. A pond has 176 fish. If 38 fish are caught and 22 new fish are added, how many fish are in the pond now? *(0 Marks)*

➤ 176 - 38 = 138, then 138 + 22 = 160 fish are in the pond now.

2. Explain why a sphere cannot be stacked like a cube, using the idea of flat and curved surfaces. *(0 Marks)*

➤ A sphere has only a curved surface and no flat faces at all, so there is no flat part for another object to rest on without rolling away, unlike a cube which has flat faces that let objects sit steadily on top of each other.

3. Write a 3-digit number where the hundreds digit is double the tens digit, and the ones digit is 5. Explain your reasoning. *(0 Marks)*

➤ If the tens digit is 1, the hundreds digit must be double that, which is 2; with the ones digit given as 5, the number is 215. Check: hundreds digit 2 is double the tens digit 1, and the ones digit is 5, so 215 fits all the conditions.

4. Two toy blocks, a cube and a cuboid, look similar from one side. Explain one way you could tell them apart just by looking at all their faces. *(0 Marks)*

➤ You can check if all six faces are exactly equal squares - if yes, it is a cube; if the faces are rectangles of different sizes (length longer than breadth or height), it is a cuboid, not a cube.

5. A number lies between 150 and 200. Its tens digit is 7 and its ones digit is the same as its tens digit. What is the number? *(0 Marks)*

➤ The number is 177 (hundreds digit 1, tens digit 7, ones digit 7, and 177 lies between 150 and 200).

6. A rectangular toy box (cuboid) is placed so its largest face touches the ground. Explain why this makes the box more stable than standing it on its smallest face. *(0 Marks)*

➤ A larger flat face touching the ground gives the box a wider base to balance on, making it less likely to tip over, while a smaller face gives a narrower, less stable base.

7. If 3 groups of 100 marbles and 4 groups of 10 marbles and 6 single marbles are combined, what is the total, and which chapter concept does this use? *(0 Marks)*

➤ 3 hundreds + 4 tens + 6 ones = 300 + 40 + 6 = 346 marbles; this uses the place value and grouping concepts from "Double Century" (extended beyond 200 as an application of the same idea).

8. A cylindrical toy roller and a spherical ball are both placed on a slope. Explain if there would be any difference in how they roll, based on their shapes. *(0 Marks)*

➤ A sphere can roll in any direction on the slope because it is round on all sides, while a cylinder will mostly roll straight along its curved side in one direction unless tipped, because it has two flat circular ends that resist rolling sideways.

9. Write the number that is 25 less than 200, and explain the subtraction steps. *(0 Marks)*

➤ 200 - 25 = 175. Steps: since 200 has 0 ones and 0 tens, borrow to make 20 tens (190 + 10), then subtract: 200 - 25 = 175.

10. A shape has 5 faces, 8 edges, and 5 corners, with one face being a square and the rest being triangles. What could this shape be, based on shapes discussed in class? *(0 Marks)*

➤ This description matches a pyramid with a square base (a square-based pyramid), which has 1 square face, 4 triangular faces, 8 edges, and 5 corners.

11. Explain why comparing only the first digit of two 3-digit numbers is not always enough to know which is greater. Give an example. *(0 Marks)*

➤ If the first (hundreds) digits are the same, we must compare further digits; for example, 152 and 149 both could be mistakenly compared using only the first digit if written differently, but properly: 152 has hundreds digit 1, tens digit 5; 149 has hundreds digit 1, tens digit 4, so comparing tens digits shows 152 is greater, proving we sometimes need to look beyond just the first digit.

12. A toy maker has 3 cubes, 2 cuboids, and 4 cylinders. If each shape has a different number of flat faces, find the total number of flat faces among all these toys (spheres and curved-only surfaces do not count). *(0 Marks)*

➤ 3 cubes have 6 faces each = 18 faces; 2 cuboids have 6 faces each = 12 faces; 4 cylinders have 2 flat faces each = 8 faces. Total = 18 + 12 + 8 = 38 flat faces.

13. A number between 100 and 200 has digits that add up to 10, and its tens digit is twice its ones digit. Find the number. *(0 Marks)*

➤ Let ones digit = x, tens digit = 2x. Hundreds digit is 1. So 1 + 2x + x = 10, meaning 3x = 9, so x = 3. Tens digit = 6, ones digit = 3. The number is 163.

14. Explain how you would check if a given box-shaped object is a perfect cube without using a ruler, just by observation. *(0 Marks)*

➤ You could compare the edges by eye or by placing the same object against each face - if all faces look like identical squares of the same size when you turn the object around, it is likely a cube; if any face looks longer or shorter than another, it is a cuboid, not a cube.

SECTION C — Short Answer Questions (Explain) (3 Marks each)

1. A trader has 200 oranges. Every day, the trader sells some oranges and this pattern continues: Day 1 sells 34, Day 2 sells 12 more than Day 1, and Day 3 sells 8 fewer than Day 2. Find the total oranges sold over the three days and the oranges left after Day 3. *(0 Marks)*

➤ Day 1: 34 oranges sold.

➤ Day 2: 34 + 12 = 46 oranges sold.

➤ Day 3: 46 - 8 = 38 oranges sold.

➤ Total sold = 34 + 46 + 38 = 118 oranges.

➤ Oranges left = 200 - 118 = 82 oranges.

2. Explain, using properties of faces, edges and corners, how you would prove to a younger student that a cube and a cuboid are related but not exactly the same shape. *(0 Marks)*

➤ Both shapes have 6 faces, 12 edges, and 8 corners, which makes them look similar in structure.

➤ However, in a cube every face is an identical square, while in a cuboid, opposite faces are equal rectangles but not necessarily squares, and not all faces need to be the same size.

➤ You could show this by comparing a dice (cube) to a brick or book (cuboid) side by side, pointing out that the dice looks the same from every direction, but the brick looks longer from one side.

3. Four numbers are hidden in a pattern: each number is 15 more than the previous one, starting from 105. Write the first four numbers in this pattern and explain if any of them cross into "Double Century" territory (200 or beyond). *(0 Marks)*

➤ First number: 105.

➤ Second number: 105 + 15 = 120.

➤ Third number: 120 + 15 = 135.

➤ Fourth number: 135 + 15 = 150.

➤ None of these four numbers reach 200, since the sequence would need several more steps of adding 15 to cross 200 (150, 165, 180, 195, 210 - so the 8th term would be the first to cross 200).

4. A model house is built using 1 cuboid (walls), 1 cone-topped structure (roof shaped like a cone on a cylinder tower), and cube-shaped decorative blocks. Explain how the different properties of these shapes (stability, rolling, pointed tops) make them suitable for their specific parts of the house. *(0 Marks)*

➤ The cuboid is used for the walls because its flat rectangular faces make it stable and easy to stack to form straight walls.

➤ The cone-topped cylinder tower works well for a roof or turret because the cone's pointed top allows rain or objects to slide off easily, while the cylinder gives height and stability underneath.

➤ The cube-shaped decorative blocks are chosen for decoration because their equal square faces make them easy to arrange in neat, even patterns.

5. A number sequence begins at 200 and decreases by 13 each time: 200, 187, 174, ... Continue the sequence for two more terms and explain the pattern of subtraction used. *(0 Marks)*

➤ Continuing: 174 - 13 = 161, and 161 - 13 = 148.

➤ So the next two terms are 161 and 148.

➤ The pattern is a constant subtraction of 13 from each previous number, which is a decreasing number pattern.

6. Explain, using examples, why understanding 3D shapes is useful beyond the maths classroom - for example, in building, packing, or everyday objects. *(0 Marks)*

➤ Builders use knowledge of cuboids and cubes to construct stable buildings, since flat-faced shapes stack and stay steady.

➤ Packers use knowledge of cuboid shapes to fit boxes tightly together in trucks or shelves without wasted space.

➤ Everyday objects like cans (cylinders), balls (spheres), and party hats (cones) are designed based on the properties of these shapes for their specific purpose, showing that shape knowledge is used all around us, not just in the classroom.

SECTION D — Long Answer Questions (5 Marks each)

1. A large box contains 200 building blocks made up of cubes, cuboids, cylinders, cones, and spheres. If there are 40 cubes, 35 cuboids, 45 cylinders, and 30 cones, find how many spheres are in the box, and then explain one property that makes spheres different from all the other four shapes in this box. *(0 Marks)*

➤ Total non-sphere blocks = 40 + 35 + 45 + 30 = 150.

➤ Number of spheres = 200 - 150 = 50 spheres.

➤ A sphere is different from the other four shapes because it has absolutely no flat faces or edges - the entire surface is curved, while cubes, cuboids, cylinders, and cones all have at least one flat face.

➤ This means a sphere can roll in every direction, while the others either cannot roll (cube, cuboid) or roll only in a limited way (cylinder, cone).

2. A number pattern game: start at 100, and each turn you can either add 15 or subtract 8. After 3 turns of adding 15 each time, what number do you reach? Then, if you subtract 8 twice from that number, what is the final number? Show every step clearly and check whether the final number is still within the "Double Century" range (up to 200). *(0 Marks)*

➤ Starting number: 100.

➤ After turn 1 (add 15): 100 + 15 = 115.

➤ After turn 2 (add 15): 115 + 15 = 130.

➤ After turn 3 (add 15): 130 + 15 = 145.

➤ After subtracting 8 once: 145 - 8 = 137.

➤ After subtracting 8 again: 137 - 8 = 129.

➤ The final number, 129, is still within the "Double Century" range since it is less than 200.

3. Design a toy shop inventory list with exactly 200 toys total, using at least four different 3D shapes (cube, cuboid, cylinder, cone, sphere), showing how many of each shape you would stock and why, then verify your numbers add up to exactly 200. *(0 Marks)*

➤ Example inventory: 45 cube-shaped building blocks, 40 cuboid-shaped toy boxes, 50 cylinder-shaped toy drums, 30 cone-shaped party hats, and 35 sphere-shaped balls.

➤ Reasoning: building blocks (cubes) and toy boxes (cuboids) are popular for construction play, so higher stock is reasonable; balls (spheres) and drums (cylinders) are popular for active play; party hats (cones) are stocked slightly less since they are seasonal.

➤ Verification: 45 + 40 + 50 + 30 + 35 = 200, which matches the total of 200 toys required.